SHIRE HIGHLANDS EDUCATION DIVISION

2024 MSCE MOCK EXAMINATIONS

CHEMISTRY - M038/I

PROVISIONAL MARKING SCHEME

PAPER I

THEORY

(100 marks)

Subject Number: M038/I

Tuesday, 26th March **Time Allowed: 2 hours**

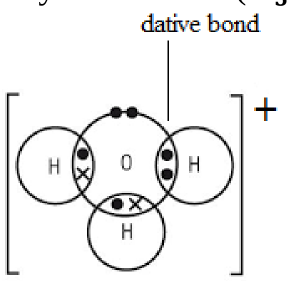
8:00-10:00 am

Instruction to teachers: *Enrich the marking schemes before you start marking*

Section A (70 marks)

1. a. (i). Dative bond is a covalent bond whereby only one atom contributes all the shared pair of electrons. (1 mark)

(ii). Structure of hydronium ion ($\boldsymbol{H_3O^+}$) and label the dative bond



1 mark for label

1 mark for correct structure

b. Hydrogen bonding / van der Waal's forces(1 mark)

c. Water has <u>more hydrogen bonding which are stronger</u> and need more energy to be broken down unlike <u>butane which has weak forces of van der Waal's forces</u>.

(2 marks)

2. a. $6.023x10^{23}$ (1 mark)

b. *Vol.* = $\frac{400\ cm^3}{100cm^3} x\ dm^3 = 0.4\ dm^3$

Mass = moles x RFM $\rightarrow \frac{0.4\ dm^3}{22.4\ dm^3mol^{-1}} x32g/mol$ *1 mark*

= 0.57g *1 mark*

c. *1 mark for any of the following*

- It shows whether the hypothesis has been supported or contradict hypothesis.
- Gives a brief summary of findings.
- Determines the success or achievement of the aim of experiment.

d. (i).

- Cooking oil (or coconut oil/vegetable oil/ animal fats/glyceryl stearate/castor oil) *1 mark*
- Concentrated sodium hydroxide *1 mark*

(ii). To reduce solubility of soap / to provide balance between softness and hardness of soap/ (1 mark)

3. a. **Electrolysis** is the decomposition of a compound which is in molten or aqueous state using electrical energy. (1 mark)

b. Any **one** of the following

Concentration/ position of the metal or ion in electrochemical series/type of electrode used. (1 mark)

c.

(i). Gases produced at electrode **A** and **B**

A oxygen gas (1 mark)

B Hydrogen gas (1 mark)

(ii). $4OH^-$ (aq) $\longrightarrow$ $2H_2O$(l) + O_2(g) + $4e^-$ (2 marks)

(iii). Iodine would be produced (1 mark)

c.

Copper lead iron aluminium magnesium or

Cu Pb Fe Al Mg (2 marks)

(i) Al^{3+} **1 mark** and Pb **1 mark**

4. a. Citric acid **1 mark**

b. CH_3COOH (aq) + H_2O (l) $\longrightarrow$ CH_3COO^- (aq) + H_3O^+ (l)

(1mark for reactants and 1 mark for products)

c. To neutralise$^{1\ mark}$ the harmful effect of acid by a base in tooth paste this stops or prevents tooth decay $^{1\ mark}$

d.

(i) Theoretical yield of carbon dioxide ($2CO_2$).

Mole ratio :3 mol $3O_2 \rightarrow$ 2 mol CO_2

Mass ratio: 6 x 16g → 2 x 44g **$^{1\ mark}$**

$$\rightarrow \frac{96g}{25} = \frac{88g}{mass\ of\ CO_2}$$ **1 mark**

Mass of CO_2 = 22.92g **$^{1\ mark}$**

(ii) percentage yield if **19.65g** of carbon dioxide was obtained.

% yield = $\frac{actual\ yield}{theoretical\ yield}\ x\ 100$

$= \frac{19.65g}{22.92g}\ x\ 100$ **$^{1\ mark}$**

= 85.73% **$^{1\ mark}$**

5. a. addition polymerisation (1 mark)

b. Terylene polymer **use** (any 1 for 1 mark)

- making clothing (trousers, shirts, golf shirts
- sails and ropes
- plastic model kits

c.

(i)

N alkane **$^{1\ mark}$**

P alkanoic acid / arboxylic acid$^{1\ mark}$

(ii) **O** **$^{1\ mark}$** (NB: identification should by letter not name)

(iii) They ionise, these ions carries charge /conduct electricity (1 mark)

d. *Any point below for 1 mark*

- Addiction (alcoholism) which can destroy physical and emotional heath.
- Causes of some cancers such as liver cancer, mouth cancer, pharynx cancer, esophagus cancer and pancreatic cancer.

- Causes cardiac problems such as high blood pressure, heart failure and stroke.
- Causes miscarriages in pregnant women.
- Can influence a person to commit suicide or suffer serious injury.
- Influences a person to indulge in risky sexual behaviours leading to unwanted pregnancies, HIV and AIDS, and other sexually transmitted infections (STIs).
- Loss of employment leading to poverty.
- Causes liver cirrhosis.

6. a.
   - Purification of metals
   - Refining gemstones
   - Identification.

   b.
   (i) **Y**: nitric acid (1 mark)

   (ii) Any 2 following (2x1 mark)
   - Very high temperature / high temperature 450 ℃- 500℃.
   - High pressure of above 250 atm.
   - Iron catalyst.

   (iii) Used in fertilizer or for fertilizer making / manufacture of herbicides and insecticides.(1 mark)

7. a. Increase in gas pressure <u>reduces spaces</u> between particles 1 mark (forces the gas particles to be close to each other) this <u>increases collision frequency</u> 1 mark of gas particles per second there by increasing rate of reaction

   b.
   (i) Magnesium sulphate 1 mark

   (ii) Rate of hydrogen reaction = $\frac{45\ cm^3 - 10\ cm^3}{7\ s} = \frac{35\ cm^3}{7s}$ 1 mark

   $= 5cm^3/s$ 1 mark

   (iii) Any 1 of the following for 1 mark
   - Hydrogen gas
   - Unused acids
   - Magnesium suphate.

8. a. Calcium hydrogen carbonate (or magnesium hydrogen carbonate) 1 mark

b.

- Evaporation/ transpiration levels get disrupted 1 mark, leading to drying up of moisture in the air and throwing off the balance of the water cycle. 1 mark
- Causes a continual dry air, low humidity and decreased precipitation 1 mark leading to drought-prone area (a desert-like climate) 1 mark

c.

- The products are filtered. 1 mark
- Precipitates are washed/cleaned using distilled water. 1 mark
- Then propanone is added to help precipitate dry quickly. 1 mark
- Precipitate are then put in oven at low temperature to dry. 1 mark

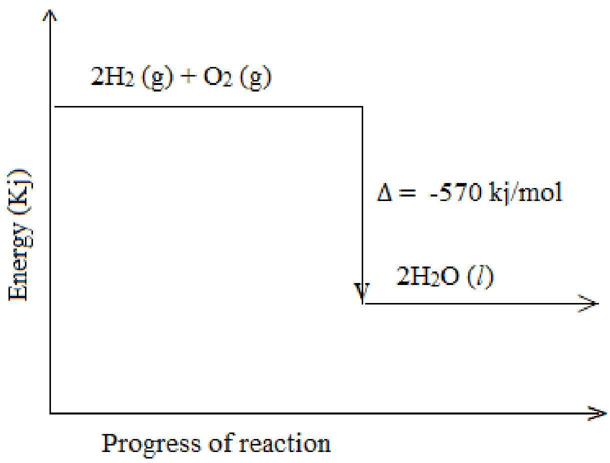
9. a. Bond energy is amount of energy required to break or form one mole of a chemical bond / a measure of bond strength in a chemical bond 1mark

b.

(i) Exothermic reaction 1mark

(ii) Gas 1mark

(iii) Energy level diagram for the reaction



1 mark for reactants

1 mark for product

10.

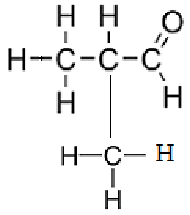
a. Butanal 1mark

b. $CH_3CH_2CH_2OH$ /$CH_3(CH_2)_3OH$ 1mark c.

(i) Hydrogen gas / H_2 1mark

(ii) Nickel metal catalyst / platinum metal catalyst.

d. **One** isomer of compound **K**



2 marks for the correct structure

1 mark for the correct name

2-methylbutanal

SECTION B (30 marks)

11. a. Describe how nitrate ion ($\mathbf{NO_3^-}$) can be tested using **brown ring experiment**.

- Add equal volume of fresh iron (II) sulphate solution and the sample suspected to contain nitrate ions. 1 mark
- Heat the mixture. 1 mark
- Hold the test tube in diagonal (slope) position, slowly and carefully add concentrated sulphuric acid. 1 mark
- If a bring ring forms 1 mark where the two layers meet, then the sample contains nitrate ions — 4 marks

b. *Any 3 points below (1 point = 1 mark its description = 1mark)*

- Creates job opportunities, people are employed to work in recycling companies, collection of raw materials form other companies and homes.
- Reduces environmental pollution, this reduces the need for convectional waste disposal mechanism such as burning and open dumping creating a pollution-free environment.
- It saves energy, reduced energy is required than in the fresh production of materials, this also helps to provide cheap but of good quality products to people.
- Helps to conserve non-renewable resources, this reduces pressure on consumption of fresh raw natural resources. (6 marks)

12. a. *Any **two** of the following. (2 x 2 marks = 4 marks)*

- Ensures constant supply of fresh water, when water evaporates from the sea, salts are left behind. This fresh water is used by plants, animals, aquatic animals and human beings.
- Maintains water balance between the water on the surface, underground and in the atmosphere.
- It provides moisture in the atmosphere through the process of evapotranspiration. The moisture helps to cool down temperature and to control humidity.
- Provides underground water which is used by plants *4 marks*

b.

- Iron oxidises when in contact with water

$$Fe\ (s) - 2e^- \xrightarrow{H_2O} Fe^{2+}\ (aq)\ \ldots..\ eq\ (i)$$ *1 mark*

- Presence of water, oxygen and electrons released in equation (i) forms hydroxyl ion (OH^-). Thus reduction *1 mark*

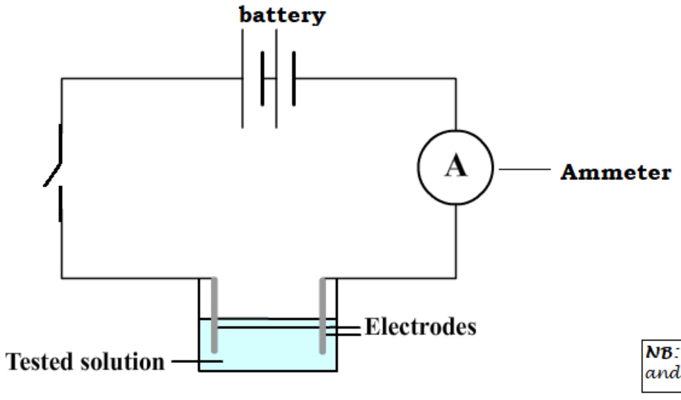
$$2H_2O\ (l) + O_2\ (g) + 4e^- \longrightarrow 4OH^-\ (aq)\ \ldots.eq\ (ii)$$ *1 mark*

- The ions Fe^{2+} and OH^- produced in equation (i) and (ii) combine to form iron hydroxide.

$$2Fe^{2+}(aq) + 4OH^-(aq) \longrightarrow 2Fe(OH)_2(s)$$ *1 mark*

- Iron hydroxide reacts with water and oxygen to form <u>rust</u>. *1 mark*

$$2Fe(OH)_2(s) + H_2O\ (l) + O_2(g) \longrightarrow Fe_2O_3xH_2O(s)$$ *1 mark*



-Battery or DC supply - 1 mark
- Ammeter - 1 mark
- Electrode/solution/beaker - 1 mark

NB: check correctness of the apparatus and name labelled

a. Set up the apparatus as shown above.
b. Dip the electrodes into the beaker containing 200ml of nitric acid. 1 mark
c. Close the switch. Note and record ammeter reading. 1 mark

Table of results

Tested solution	**Ammeter reading (A)**
Nitric acid (HNO_3)	
Ethanoic acid (CH_3COOH)	

1 mark for table of result

d. Remove the electrodes from the beaker and clean them with distilled water, then rub them with a dry clean cloth. 1 mark
e. Repeat steps b to d using a beaker of 200ml ethanoic acid. 1 mark
f. Compare the ammeter reading
The acid that gives a higher ammeter reading is a strong acid 1 mark because it dissociate completely in water releasing many ions which conducts more current 1 mark unlike a weak acid which partially dissociates and releases few ions which conducts less current. (10 marks)

THE END OF PROVISIONAL MARKING SCHEME